A Biomimetic Vision-Guided Robot Modeling Cuttlefish Cognition in a Dynamic Foraging Environment

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Introduction

Project Goal:

- Create a biomimetic robot, the CuttleBot, that embodies the cognitive and behavioral abilities of cuttlefish
- Leverage the expertise of a multidisciplinary team

Background:

- Recent research highlights the cuttlefish's advanced cognitive skills and its excellent hunting and defending behaviors

Contributions:

- **Camouflage:** Camouflage with its environment by capturing the textures of its surroundings
- **Stereopsis and Tentacle Strike:** Find distance of prey using the disparity between its two cameras to perform an attack with its claw
- **Episodic Memory Model:** Use goal-directed reinforcement learning to store and recall predator and prey events

Materials and Methods

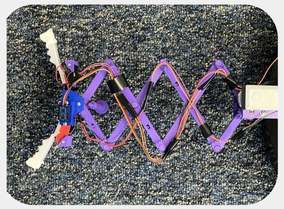
Hardware Configuration and Control:

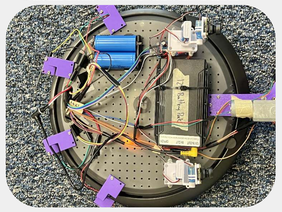
- **Robotic Base and Control:** The CuttleBot is built on a TurtleBot4 Lite, which uses a Raspberry Pi 4B for central processing. The robot communicates to an NVIDIA Jetson Orin for behavioral selection and visual processing
- **Visual Processing:** Two high-resolution, wide-angle cameras mounted on a dual-axis pan-tilt unit, providing a 175-degree field of view

Mechanical Abilities:

- **Extendable/Retractable Claw:** Scissor mechanism designed to perform long-range prey capture using a motor for extension and retraction and a servo for grabbing and releasing
- **Adaptive Display Shell:** Custom-designed, 3D-printed semi-transparent shell integrated with programmable LED matrices for rapid color and pattern changes, simulating cuttlefish skin camouflage and communication signals





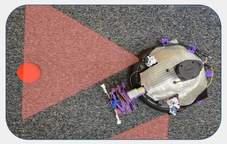


Methods

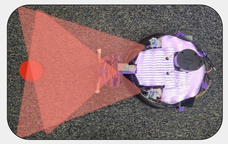
Behavioral Sequences:

Attack Phase: Robot employs object recognition for alignment before striking, with a hypnotic pattern to distract and capture prey

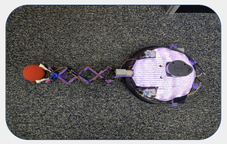
Roaming, Sees prey



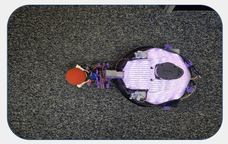
Aligns and hypnotizes



Extends tentacles

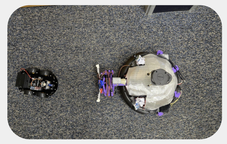


Retracts prey and eats

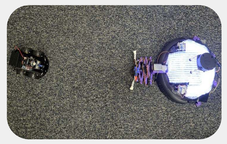


Defense Phase: Robot employs defense mechanisms of blanching (a sudden loss of color used to startle predators), retreating, and camouflaging

Roaming, Sees predator



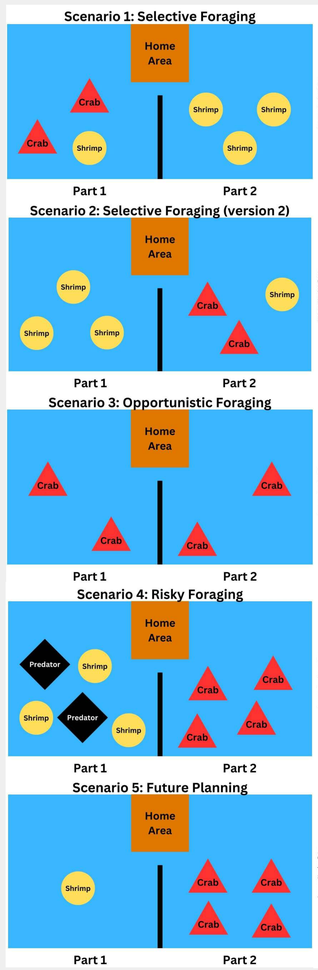
Blanches and backs up



Retreats and camouflages



Experimental Environment and Simulation Graphs:



Scenario 1: Selective Foraging

- **Visual:** More desirable prey (shrimp) located in Part 2; less desirable prey (crab) in Part 1
- **Graph:** Over 15 days, CuttleBot increasingly prioritized visits to Part 2, indicating a preference-based foraging strategy

Scenario 2: Selective Foraging 2

- **Visual:** Scenario reversed, shrimp in Part 1, crab in Part 2
- **Graph:** CuttleBot adapted by shifting its foraging to Part 1 over 15 days, again favoring the more rewarding prey

Scenario 3: Opportunistic Foraging

- **Visual:** Only less desirable prey (crab) present in both parts
- **Graph:** CuttleBot showed no preference, distributing visits equally between both parts, indicating undirected or neutral foraging when options are equivalent

Scenario 4: Risky & Selective Foraging

- **Visual:** Part 1 offers shrimp but includes predators; Part 2 contains only crabs and is safe
- **Graph:** CuttleBot balanced risk and reward by visiting both parts evenly while successfully avoiding predators

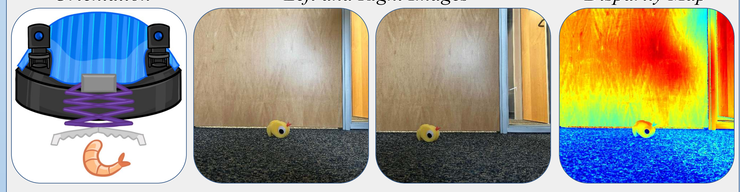
Scenario 5: Future Planning

- **Visual:** Part 1 contains intermittently appearing shrimp; Part 2 has a constant supply of crabs
- **Graph:** CuttleBot slightly learned to anticipate the appearance of shrimp, but still showed some selective foraging for crabs

Results and Discussion

- **Parallel Focused Stereopsis:** The CuttleBot centers the prey in its field of view, using the differences in imagery to ascertain depth

Orientation | *Left and Right Images* | *Disparity Map*



- **Object Focused Stereopsis:** The CuttleBot centers each camera on the prey, using the differences in the foreground and background to ascertain depth

Orientation | *Left and Right Images* | *Disparity Map*



- **Texture-Based Camouflage:** The CuttleBot is able to capture the texture of its environment better replicating the cuttlefish's abilities



- **Episodic Memory Model Supported Adaptive Behavior:** By recalling past predator-prey events, the model supported flexible, goal-directed actions, key for avoiding risk (Scenario 4) and planning ahead (Scenario 5)
- **Future Steps:** Integrate SLAM for spatial memory, build a soft biomimetic claw, improve camouflage via high-res textures, and add polarized vision for realistic perception

References

Billard, P., Schnell, A.K., Clayton, N.S., Jozet-Alves, C.: Cuttlefish show flexible and future-dependent foraging cognition. Biology Letters 16 (2020). https://doi.org/10.1098/rsbl.2019.0743

Birch, J., Schnell, A.K., Clayton, N.S.: Dimensions of animal consciousness. Trends in Cognitive Sciences 24(10), 789–801 (2020). https://doi.org/10.1016/j.tics.2020.07.007

Feord, R.C., Sumner, M.E., Pusdekar, S., Kalra, L., Gonzalez-Bellido, P.T., Wardill, T.J.: Cuttlefish use stereopsis to strike at prey. Science Advances 6(2), eaay6036 (2020). https://www.science.org/doi/10.1126/sciadv.aay6036

Hanlon, R.T., Messenger., J.B.: Cephalopod Behaviour. Cambridge University Press, second edn. (2018)

Hwu, T., Krichmar, J.: Neurorobotics: Connecting the Brain, Body and Environment. MIT Press, Cambridge, MA (2022)

Jozet-Alves, C., Bertin, M., Clayton, N.S.: Cephalopod cognition: Complex learning and memory in a flexible invertebrate brain. Current Biology 23(23), R1033–R1035 (2013)

Schnell, A.K., Clayton, N.S.: Cephalopods: Ambassadors for rethinking cognition. Biochemical and Biophysical Research Communications 564, 27–36 (2021). https://doi.org/https://doi.org/10.1016/j.bbrc.2020.12.062

Schnell, A.K., Clayton, N.S., Hanlon, R.T.: Cuttlefish exert self-control in a delay of gratification task. The Royal Society: Biological Sciences 288 (2021). https://doi.org/http://doi.org/10.1098/rspb.2020.3161